# Intelligent Transportation Systems and Traffic Operations

## Introduction

Instructor

Heeseung Shon, Ph.D.

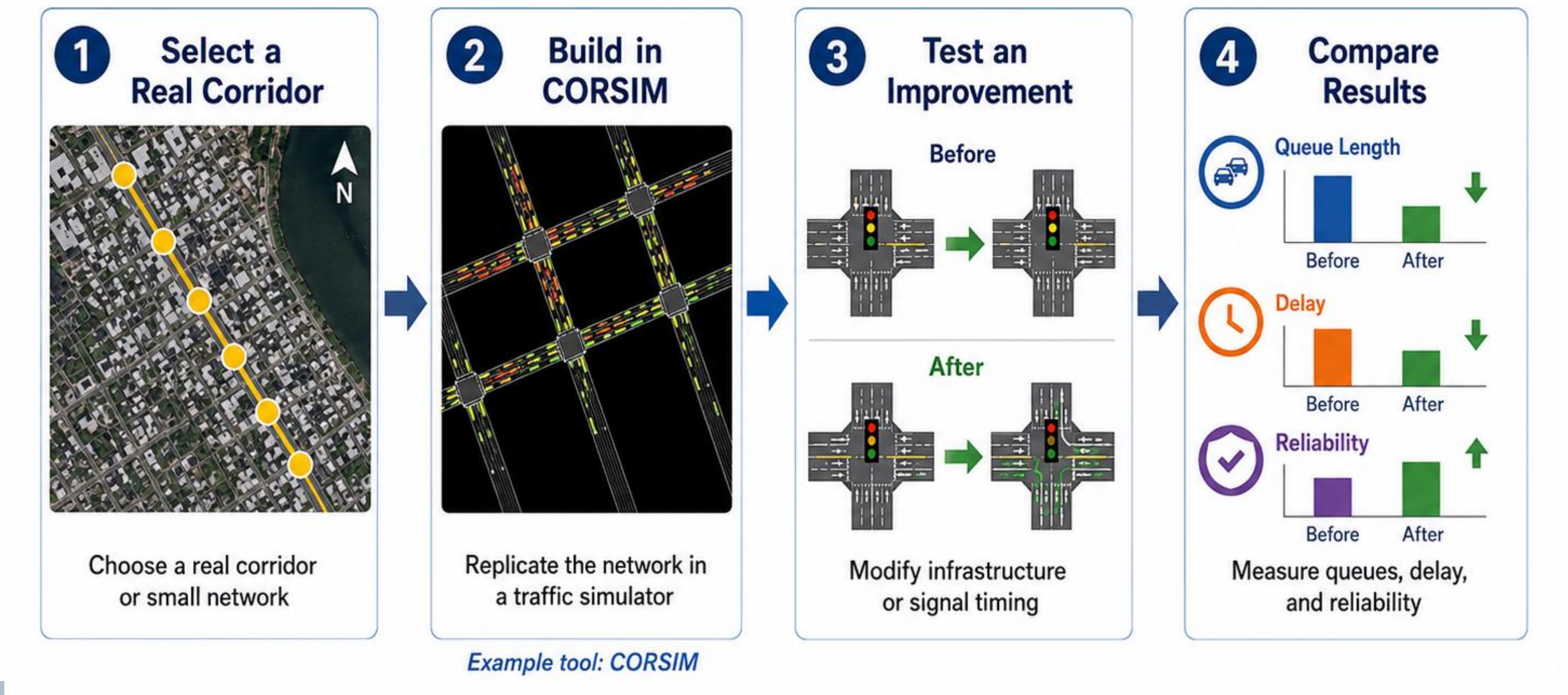
# Course Objectives

- **This course aims to develop students' knowledge and skills in:**

  - **Foundations of transportation engineering**: construct and read time-space diagrams; analyze queues, congestion, and capacity using queueing and traffic-flow theory; design and evaluate traffic signal timing.
  - **Network modeling and trip assignment**: predict travel demand with the four-step model; perform and interpret user-equilibrium and system-optimal assignments; explain network paradoxes.
  - **Smart mobility and emerging technologies:** Explain next-generation transportation; explore AI methods for predicting traffic speed and flow; assess the sustainability of transportation modes.

# Mini Project

❖ Select a real corridor or small network; replicate it in a traffic simulator; implement an infrastructure change or a signal plan to improve traffic conditions; document changes in queues, delay, and reliability.



# Textbook and References

- Lecture Note

- Principles of Highway Engineering and Traffic Analysis (Mannering and Washburn), Chapters 5-8.

# Class Contents

What is Transportation Engineering?

Time-Space Diagram

Traffic Properties & Measurement

Queuing Theory & Analysis

Traffic Flow Theory

Signal System (Cyclic System)

4-Step Model

Trip Assignment

Network Paradoxes

Cooperative-ITS (C-ITS)

Future Transportation Modes

# Intelligent Transportation Systems and Traffic Operations

## What is Transportation Engineering?

**Instructor**

**Heeseung Shon, Ph.D.**

# What is Transportation Engineering?

## Overview

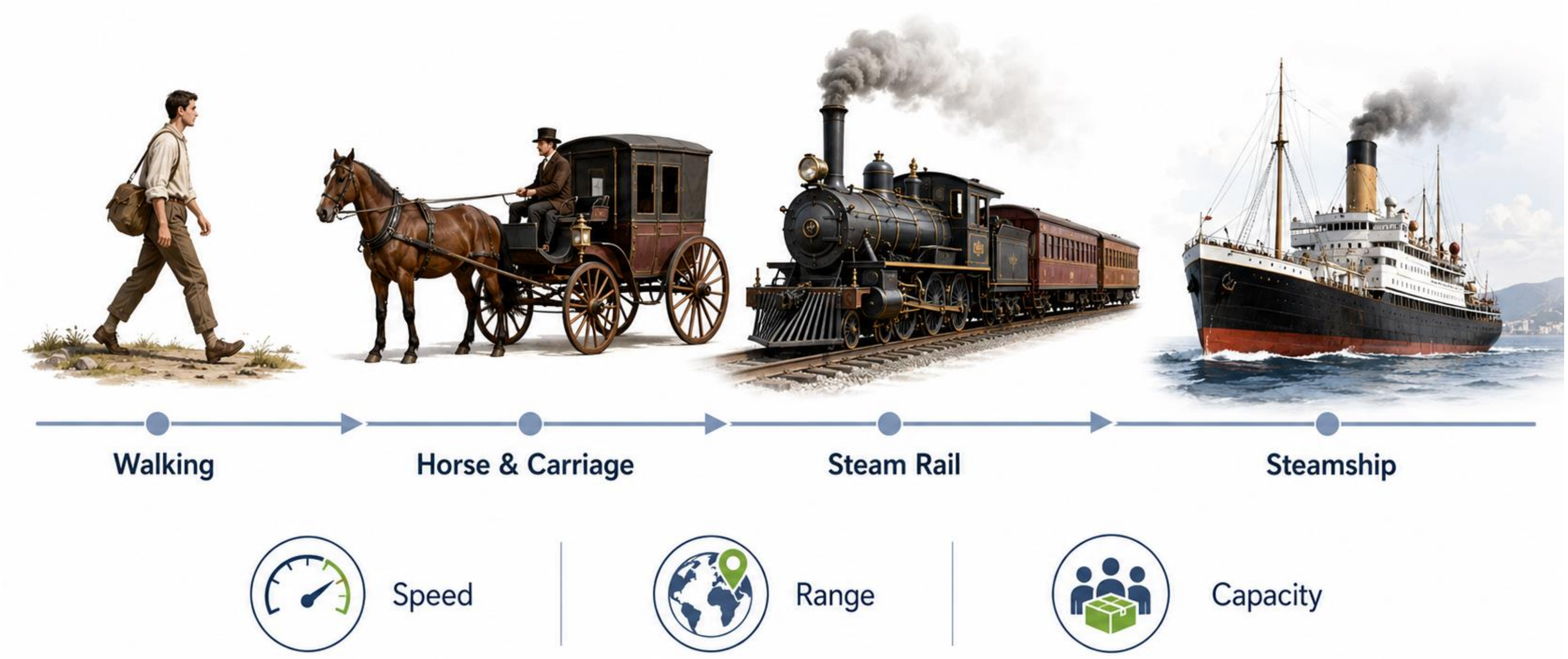
### ❑ Definition

- All the activities involved in carrying, moving or conveying either **people** or **goods**

# What is Transportation Engineering?

## History of Transportation

- ❑ **From human power to mechanized transportation**
  - ➢ Transportation gradually reduced the constraints of **distance** and **time**.

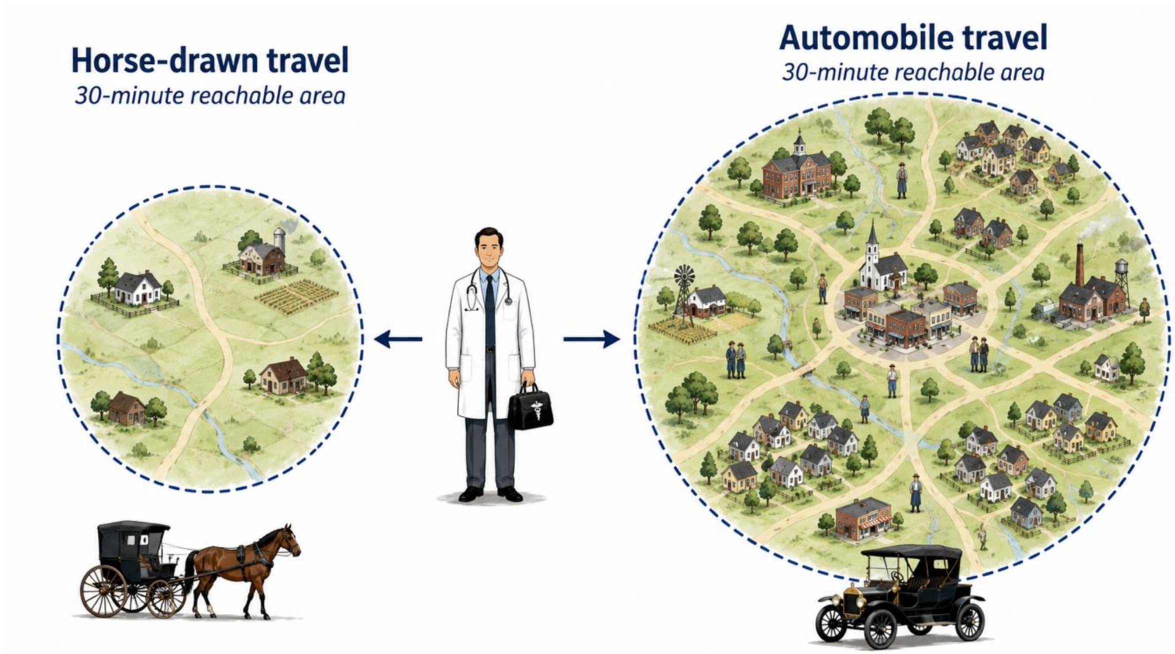


# What is Transportation Engineering?

## History of Transportation

### ❑ The Automobile: Expanding Access to Opportunities

- ➢ More destinations could be reached within the same amount of time.
- ➢ Access to employment, education, markets, and healthcare expanded.

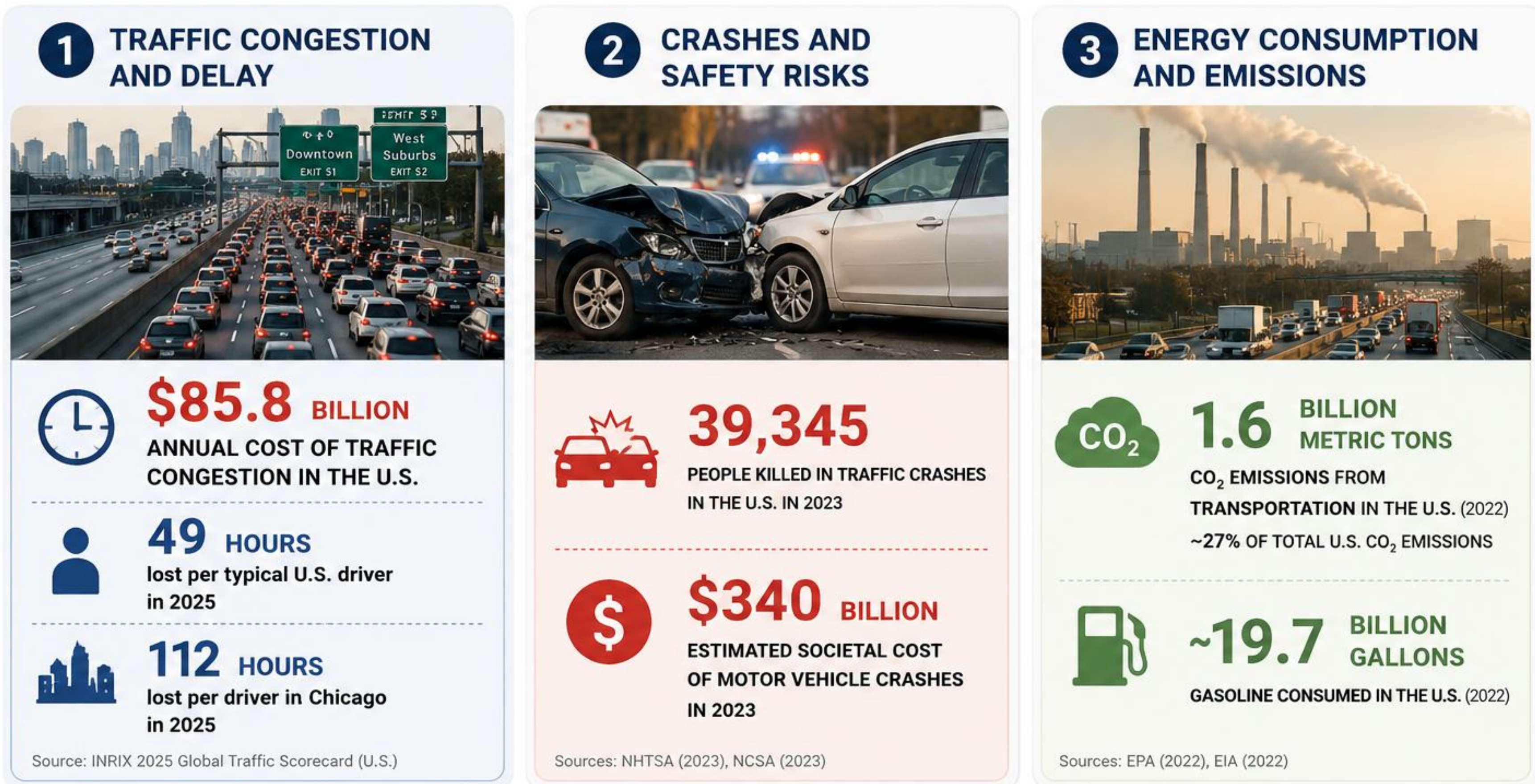



The value of transportation is not movement itself, but **access** to **opportunities**.

# What is Transportation Engineering?

## History of Transportation

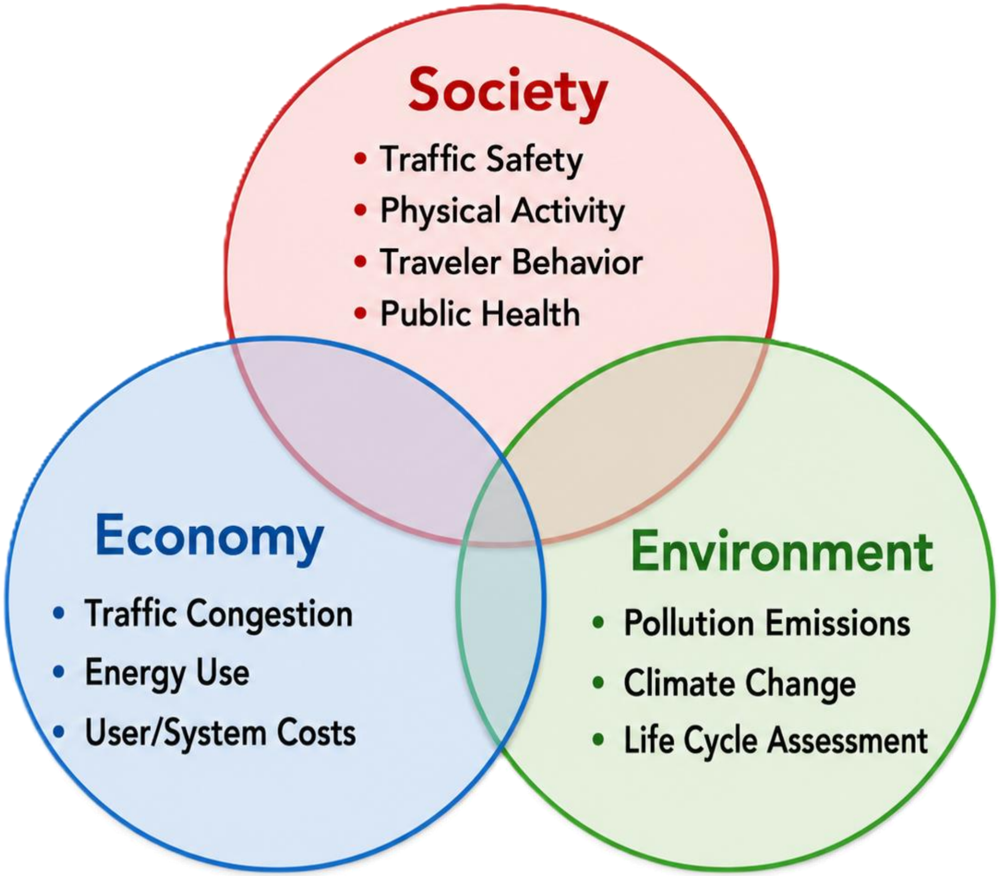
### ❑ Transportation Challenges



# What is Transportation Engineering?

## History of Transportation

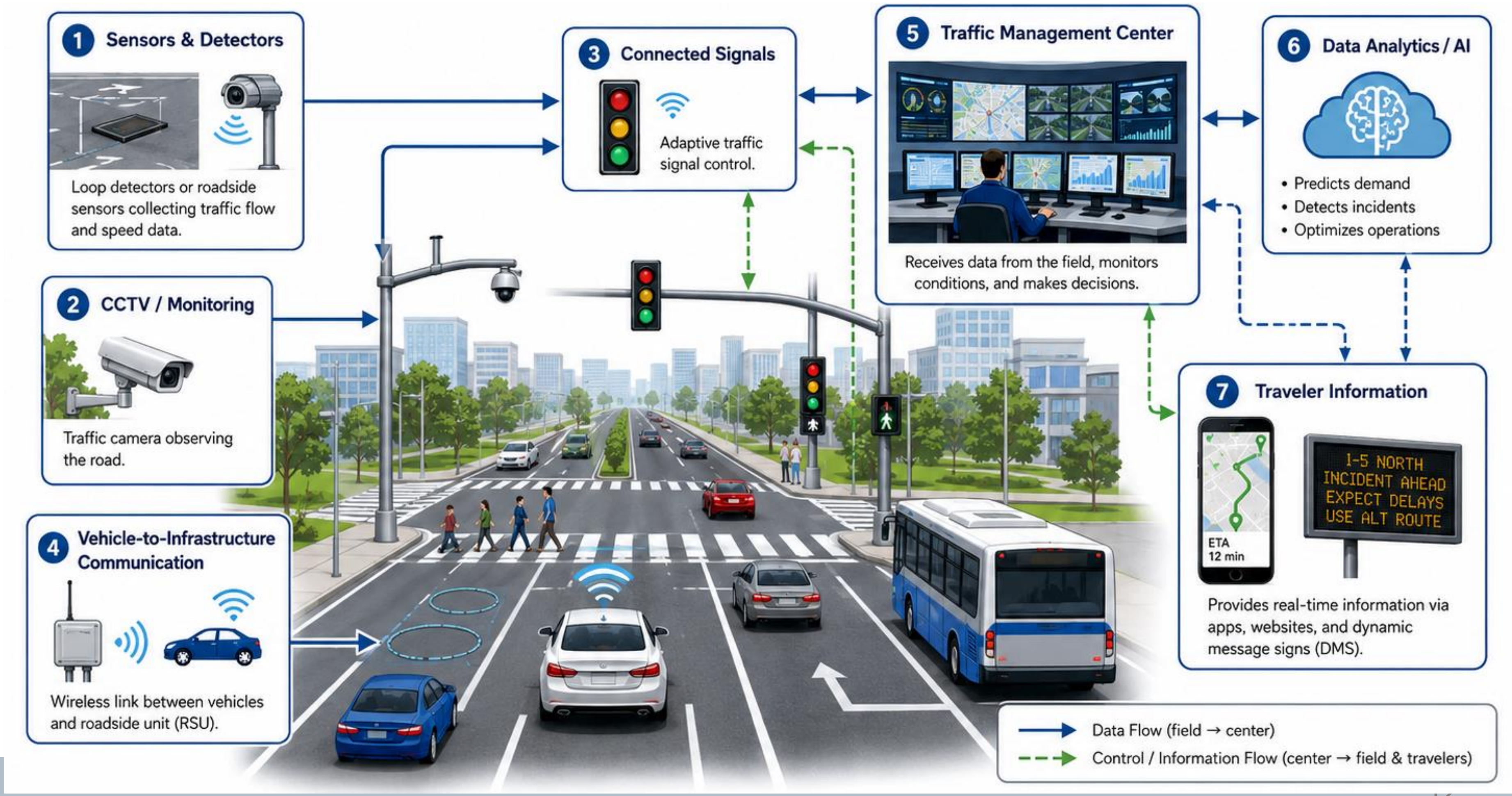
- ❑ **Sustainable Transportation**



# What is Transportation Engineering?

## History of Transportation

❑ **Intelligent Transportation Systems (ITS)**



# What is Transportation Engineering?

## History of Transportation

- **Five primary functional area of ITS**
  - Advanced Traffic Management Systems (ATMS)
  - Advanced Traveler Information Systems (ATIS)
  - Commercial Vehicle Operations (CVO)
  - Advanced Public Transportation Systems (APTS)
  - Advanced Vehicle Control Systems (AVCS)

# What is Transportation Engineering?

## Future Transportation

- ❑ **Cooperative-ITS (C-ITS)**

# What is Transportation Engineering?

## Future Transportation

- **Advanced Air Mobility Systems**
  - Electric Vertical Take-Off and Landing (eVTOL)

eVTOL

Vertiport

Sources: Joby Aviation

# What is Transportation Engineering?

## Future Transportation

❑ **Advanced Air Mobility Systems**

Port of Doha → Katara Cultural Village
30 min by car → 8 min by eVTOL
Human-carrying pilotless trial flight, 2025

Sources: EHang, 2025